

Some preliminary exercises

Q. 1 [Due March 31]

Suppose that $p(x, y)$, the joint probability mass function of X and Y , is given by

$$p(0, 0) = 0.4, \quad p(0, 1) = 0.2, \quad p(1, 0) = 0.1, \quad p(1, 1) = 0.3.$$

1. Calculate the probability mass function of X .
2. Calculate the probability mass function of Y .
3. Calculate the probability mass function of X given $Y = 0$.
4. Calculate the probability mass function of X given $Y = 1$.
5. Calculate the probability mass function of Y given $X = 0$.
6. Calculate the probability mass function of Y given $X = 1$.

Q. 2 [Due March 31]

Suppose that $f(x, y)$, the joint density of X and Y , is given by

$$f(x, y) = \begin{cases} \frac{12}{5}x(2 - x - y) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

1. Calculate the probability density function of X .
2. Calculate the probability density function of Y .
3. Calculate the conditional probability density function of X given Y .
4. Calculate the conditional probability density function of Y given X .
5. Find $\mathbb{P}(X > 0.5 \mid Y = 0.5)$.
6. Find $\mathbb{P}(X > 0.5 \mid Y > 0.5)$.

Simple Symmetric Random Walk

Q. 3 [Due Monday April 28]

Consider the *random experiment* of tossing a fair coin 10 times.

1. In a random experiment what is meant by an *outcome*? Write an example of an outcome of the above random experiment.
2. In a random experiment what is meant by the *sample space*? In the above random experiment what is the sample space?
3. How many elements are in the sample space of the above random experiment?
4. In the context of random experiments what is meant by an *event*? Provide an example of an event in the above random experiment.

Suppose a fair coin is tossed 1000 times. Each time the coin lands on heads you win 1€, if the coin lands on tails you lose 1€.

5. What is the probability that after 1000 tosses your net gain is at least 30€? Do this in two ways: (1) use the Binomial distribution; (2) use Normal approximation (central limit theorem.) You might need to use a mathematical software.
6. Find x such that the probability that after 1000 tosses your net gain is at least x € is 0.9. As before do this in two ways, once using the Binomial distribution, and once using Normal approximation.

Suppose a fair coin is tossed 1000 times. Each time the coin lands on heads you win 2€, if the coin lands on tails you lose 1€.

7. What is the probability that after 1000 tosses your net gain is at least 500€?
8. Find x such that the probability that after 1000 tosses your net gain is at least x € is 0.9.

Q. 4 [Due Monday April 28]

Consider sample paths of the one-dimensional simple random walk.

1. How many paths are there from $(0, 0)$ to $(20, 6)$?
2. How many paths are there from $(0, 0)$ to $(20, 6)$ which touch or cross the line $Y = -1$?
3. How many paths are there from $(0, 0)$ to $(20, 6)$ which do not go below the X -axis?
4. How many paths are there from $(0, 0)$ to $(20, 6)$ which do not return to a point on the X -axis?
5. How many paths are there from $(0, 0)$ to $(20, 6)$ which do not touch or cross the line $Y = 8$?

Simple Asymmetric Random Walk

Q. 5 [Due Monday May 26]

Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$X_1 = \begin{cases} 1 & \text{with probability } 0.4 \\ -1 & \text{with probability } 0.6 \end{cases}.$$

Let $S_0 \equiv 0$ and $S_n = X_1 + \dots + X_n$ for $n = 1, 2, 3, \dots$. What is the probability that the random walk $(S_n)_{n=0}^\infty$ -

1. reaches -6 ?
2. reaches 5 , and does not reach -6 before reaching 5 ?
3. reaches -6 and does not reach 5 before reaching -6 ?
4. reaches 5 ?
5. reaches 5 , and reaches -6 before reaching 5 ?

Order Statistics

Q. 6 [Due Monday May 26]

Let $X_1, \dots, X_n$ be i.i.d. Exponential(1) random variables. Let $X_{(1)} := \min\{X_1, \dots, X_n\}$ denote the and $X_{(n)} := \max\{X_1, \dots, X_n\}$.

1. Find the probability distribution function and the probability density function of $X_{(1)}$.
2. Find the probability distribution function and the probability density function of $X_{(n)}$.

Borel-Cantelli Lemmas

Q. 7 [Due Monday May 26]

Let $(A_n)_{n=1}^\infty$ be a sequence of events. Recall

$$\limsup A_n = \bigcap_{m=1}^\infty \bigcup_{n=m}^\infty A_n = \{A_n \text{ happens for infinitely many } n\},$$

and

$$\liminf A_n = \bigcup_{m=1}^\infty \bigcap_{n=m}^\infty A_n = \{A_n \text{ happens for all but finitely many } n\}.$$

Show that $\mathbb{P}(\limsup A_n) \geq \limsup \mathbb{P}(A_n)$, and $\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n)$.

Modes of Convergence

Q. 8 [Due Monday May 26]

Let X_1, X_2, X_3 be a sequence of random variables which converge in distribution to a constant random variable. Show that the convergence also occurs in probability.